Satvik Singh

Postdoctoral Researcher | Technical University of Munich

Department of Mathematics

satvik.singh@tum.de | Google Scholar

CURRICULUM VITAE

October 2026

RESEARCH PROFILE

Quantum information theorist studying the limits of information storage and transmission in noisy quantum devices. Research spans channel capacities, quantum error correction and entanglement theory, with current work on task-specific communication and hybrid classical–quantum memories.

ACADEMIC APPOINTMENTS AND EDUCATION

Sep 2025–present
Postdoctoral Researcher
Technical University of Munich, Department of Mathematics
Working with Prof. Michael Wolf and Prof. Robert König.

Oct 2021–Jul 2025
PhD, University of Cambridge
Department of Applied Mathematics and Theoretical Physics (DAMTP)
Supervisor: Prof. Nilanjana Datta
Thesis: *Detection and estimation of quantum channel capacities.*

Aug 2016–Jun 2021
BS–MS dual degree, IISER Mohali
Indian Institute of Science Education and Research, Mohali
Cumulative GPA: **9.9/10**
Thesis: *On the uniqueness of canonical commutation relations in quantum physics.*

HONOURS, AWARDS AND SCHOLARSHIPS

2026
Salje Medal for Science, Clare Hall
Annual college award recognising an outstanding PhD thesis in the sciences at the University of Cambridge.

2025
Václav Votruba Prize
Best thesis in theoretical physics, recognising original advances in quantum channel capacities and the conditions for reliable quantum communication.

2024
Smith-Knight/Rayleigh-Knight Prize
Grade 1 for an outstanding research essay at DAMTP, University of Cambridge, containing significant original research results.

2021–2025
Cambridge Trust International Scholarship
Funding for doctoral studies at the University of Cambridge.

2016–2021
INSPIRE Scholarship
Government of India funding for bachelor's and master's studies in natural sciences.

2016–2021
Certificates of Academic Excellence, IISER Mohali
Perfect scores of 10/10 in semesters I, II, III, IV, VI and VII.

2020
70th Lindau Nobel Laureate Meeting
Selected to represent India.

2017
Asian Science Camp
Selected to represent India.

PEER-REVIEWED PUBLICATIONS

14 articles. Listed by publication year, most recent first.

01 **Information storage and transmission under Markovian noise**
S. Singh and N. Datta
PRX Quantum **7**, 020312 (2026). | arXiv:2504.10436

02 **Entanglement in cyclic sign invariant quantum states**
A. Gulati, I. Nechita and **S. Singh**
Journal of Mathematical Physics **66**, 122202 (2025). | arXiv:2501.04786

03 **Zero-error communication under discrete-time Markovian dynamics**
S. Singh, M. Rahaman and N. Datta
Quantum **9**, 1910 (2025). | arXiv:2402.18703

04 **Simultaneous superadditivity of the direct and complementary channel capacities**
S. Singh and S. Strelchuk
IEEE Transactions on Information Theory **71(11)**, 8542–8550 (2025). | arXiv:2301.05142

05 **Ergodic theory of diagonal orthogonal covariant quantum channels**
S. Singh, N. Datta and I. Nechita
Letters in Mathematical Physics **114**, 121 (2024). | arXiv:2206.01145

06 **Fully undistillable quantum states are separable**
S. Singh and N. Datta
IEEE Transactions on Information Theory **69(10)**, 6564–6570 (2023). | arXiv:2207.05193

07 **A brief discourse on Hausdorff dimension and self-similarity**
S. Singh
The American Mathematical Monthly **129(9)**, 831–845 (2022). | arXiv:2012.10606

08 **Coherent information of a quantum channel or its complement is generically positive**
S. Singh and N. Datta
Quantum **6**, 775 (2022). | arXiv:2107.00392

09 **Diagonal unitary and orthogonal symmetries in quantum theory: II. Evolution operators**
S. Singh and I. Nechita
Journal of Physics A: Mathematical and Theoretical **55**, 255302 (2022). | arXiv:2112.11123

10 **The PPT² conjecture holds for all Choi-type maps**
S. Singh and I. Nechita
Annales Henri Poincaré **23**, 3311–3329 (2022). | arXiv:2011.03809

11 **Detecting positive quantum capacities of quantum channels**
S. Singh and N. Datta
npj Quantum Information **8**, 50 (2022). | arXiv:2105.06327

12 **Diagonal unitary and orthogonal symmetries in quantum theory**
S. Singh and I. Nechita
Quantum **5**, 519 (2021). | arXiv:2010.07898

13 **Entanglement detection in triangle-free quantum states**
S. Singh
Physical Review A **103**, 032436 (2021). | arXiv:2010.11891

14 **A graphical calculus for integration over random diagonal unitary matrices**
I. Nechita and **S. Singh**
Linear Algebra and its Applications **613**, 46–86 (2021). | arXiv:2007.11219

PREPRINTS

6 preprints. Listed by first posting, most recent first.

15 **Transmitting algebras through quantum channels**
R. Salzmann and **S. Singh**
Preprint (2026). | arXiv:2609.40054

16 **The entanglement-assisted transmission capacity is a strong converse bound for identification**
S. Singh
Preprint (2026). | arXiv:2608.11000

17 **A short proof of the modified Kretschmann–Schlingemann–Werner conjecture**
S. Singh
Preprint (2026). | arXiv:2606.16418

18 **Gaussian mean width strong converse bound on the classical identification capacity of quantum channels**
S. Singh
Preprint (2026). | arXiv:2606.05032

19 **Discriminating idempotent quantum channels**
S. Singh and B. Bergh
Preprint (2026). | arXiv:2603.28582

20 **Bi-PPT channels are entanglement breaking**
A. Müller-Hermes and **S. Singh**
Preprint (2022; revised 2023). | arXiv:2204.01685

TEACHING AND RESEARCH SUPERVISION

2021–2025 **Part II Quantum Information and Computation**
Exercise-class tutor, University of Cambridge.
Michaelmas 2021; Lent 2023, 2024 and 2025.

Cambridge **Master's thesis supervision (Part III)**
Supervised 5 projects: Alexander Makarovskiy, Amier Hussen, Jake Intrater, Chaitanya Gupta and Dibyendu Roy.

Cambridge **Undergraduate research supervision (Part II)**
Supervised 1 project: Anna Luchnikov.

TALKS AND SEMINARS

Apr 2026 **Laboratoire de Physique Théorique, Toulouse** (seminar)
Information storage under Markovian noise

Jan 2026 **Doppler Institute, Prague** (invited prize lecture)
Information storage under Markovian noise

Mar 2025 **Department of Mathematics, University of Illinois Urbana-Champaign** (seminar)
Information transmission under Markovian noise

Feb 2025 **Quantum Information Processing (QIP), Raleigh** (contributed)
Information transmission under Markovian noise

Feb 2025 **QLunch, University of Copenhagen** (seminar)
Information transmission under Markovian noise

TALKS AND SEMINARS (CONTINUED)

Jul 2024 — **Random Quantum Channels: Entanglement and Entropies, Marseille** (invited)
Zero-error communication under discrete-time Markovian dynamics

Mar 2024 — **Qinfo group, ENS Lyon** (seminar)
Zero-error communication, scrambling, and ergodicity

Oct 2023 — **Department of Mathematics, University of Oslo** (seminar)
The PPT² conjecture holds for all Choi-type maps

May 2023 — **Mathematical Physics in Quantum Technology, Edinburgh** (invited)
Diagonal unitary covariant quantum channels

Sep 2022 — **Beyond IID in Information Theory, Shenzhen/online** (contributed)
Fully undistillable quantum states are separable

Jul 2022 — **ESQuisses Summer School, Porquerolles** (invited)
Detecting positive quantum capacities of quantum channels

Mar 2022 — **Quantum Information Processing (QIP), Pasadena** (contributed)
Detecting positive quantum capacities of quantum channels

Sep 2021 — **Asian Quantum Information Science (AQIS), Tokyo/online** (contributed)
Detecting positive quantum capacities of quantum channels

Sep 2021 — **Beyond IID in Information Theory, online** (contributed)
Detecting positive quantum capacities of quantum channels

Feb 2021 — **Quantum Information Processing (QIP), Munich/online** (contributed)
The PPT² conjecture holds for all Choi-type maps

Dec 2020 — **Bringing Young Mathematicians Together (BYMAT), online** (contributed)
A graphical calculus for integration over random diagonal unitary matrices

REFEREES

Prof. Nilanjana Datta
University of Cambridge
n.datta@damtp.cam.ac.uk

Dr. Ion Nechita
Laboratoire de Physique Théorique, Toulouse
ion.nechita@univ-tlse3.fr